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Reading Guide

what you are seeing on the cover, and how to use it

What is this? A live, interactive visualization showing that the same set of integers can be measured two completely different ways. On one side, integers form a familiar geometric shape (here a hexagon, since we are showing hexagonal numbers). On the other, the same integers are arranged on a finite p -adic residue tree, where two numbers are 'close' when their difference contains many factors of the prime p . The slider lets you watch one view dissolve smoothly into the other.

WHAT IS SHOWN ON THE COVER

The cover is a real screenshot. Settings: **Hexagonal** family, $N = 28 = H_4$, $p = 3$, $d = 3$, **style = thread**, with sample pair $a = 15$, $b = 18$. The thick red line connecting 15 and 18 visually answers: 'how far apart are these integers under the p -adic metric?' The reckoner panel reports $|15 - 18|_3 = 1/3$, and 15 is congruent to 18 modulo 3, so they share one tree-branch from the root.

EVERY FEATURE, IN ORDER

1 PILLAR I MARKER

The small label at top-left that reads **PILLAR I**. Tap or click it to open a pop-up dialog explaining what Pillar I is and how it fits into the planned seven-pillar exposition.

2 DUAL MARKER

The vermilion italic word **Dual** in the title. Tap to open a second pop-up explaining what 'dual' means here: two complementary ways of measuring the same integers, the Euclidean face and the p -adic face.

3 CARDINALITY (N)

How many integers to display. Click the dropdown to choose. Options shown are family-specific values: $H_3 = 15$, $H_4 = 28$, $H_5 = 45$, etc.

4 HIGHLIGHT FAMILY

Which figurate family to color in vermilion. Five options: triangular, pentagonal, hexagonal, centered hexagonal, and star. Each has a different geometric layout and a different residue distribution mod p .

5 TREE DEPTH (D)

How many levels deep to draw the p -adic tree. Higher d means more leaves (p^d total) and finer-grained equivalence classes. Range: 1 to 4.

6 FAMILY FORMULA

The closed-form expression for the active family. For hexagonal: $H_k = k(2k - 1)$. Read-only; updates with the family.

7 PRIME (P)

The prime that defines the p -adic metric. Available: 2, 3, 5, 7. Different primes group integers differently. For example, 12 and 15 are p -adic close for $p = 3$ but distant for $p = 2$.

8 CONNECTOR STYLE

How the dashed family-lineage threads are drawn between vermilion dots. **Shells** traces concentric layers; **thread** draws a single winding path through the family members in order.

9 AUTO-MORPH (PLAY / PAUSE)

Tap to play or pause continuous morphing. When playing, the icon switches to a pause symbol and the label reads 'pause morph'. The slider eases back and forth between the Euclidean and tree views automatically.

10 CAPTURE: PNG + JSON

Tap to download two paired files: a high-resolution PNG of exactly what is on screen, and a JSON manifest recording every parameter, every dot's position and leaf assignment, every residue class, plus a SHA-256 hash for integrity verification. Both files share the same hash prefix.

11 CHIPS A AND B

Two slots holding the currently selected pair. Tap a populated chip to clear that slot. Tap any dot in the figure to assign it to the next open slot.

12 TAP DOTS TO SELECT

Click or tap any dot in the canvas. The first selection becomes a , the second becomes b , and the distance reckoner panel below fills in with every distance computation for that pair. On the cover, dots 15 and 18 are already selected.

13 MORPH SLIDER: BEGIN

The left edge of the slider track. **BEGIN** means the figure shows the Euclidean dot-figure: the integers laid out in the plane forming the family's characteristic geometric shape.

14 SLIDER TRACK: DRAG TO MORPH

Drag the slider ball along the track to morph the figure. Anywhere along the track is a smooth interpolation between the two views. You can also use auto-morph (#9) to play it back and forth automatically.

15 MORPH SLIDER: END

The right edge of the slider track. **END** means the figure shows the Bruhat-Tits tree T_p : the same integers arranged on the p -adic residue tree.

16 DISTANCE RECKONER

The left card under the figure. When two dots are selected, this fills in with the integers, $|a - b|$, $v_p(a - b)$ (the p -adic valuation), $|a - b|_p$ (the p -adic distance), the tree-meet level, and the congruence statement.

17 STRUCTURAL NOTE

The right card. A short prose summary of what the family does in residue space at the current p and d : which residue classes are inhabited, which are 'barren' (never visited), and how that pattern shifts with depth.

18 FOOTER

Version stamp, build identifier, copyright, license, and a parameters line recording the exact configuration. Anyone can reproduce the figure from the footer alone.

19 ONE LAST THING

If you read this far: notice the bottom row of features. Numbers 7, 13, 19 are all primes, and #7 is where the artifact's own prime parameter lives. We hope you enjoy playing with the tool. There is a lot more to find.

WHAT THE EXPORTS GIVE YOU

The PNG is a high-resolution snapshot at 3x scale (about 3840 by 2200 pixels), ready for print or screen. Its provenance band records version, build tag, and full parameters. **The JSON** is a structured manifest: every dot's value, leaf index, and base- p digits; the residue distribution at mod p and mod p^2 ; the selected pair's full distance computation; and a SHA-256 hash of the canonicalized export. The first 12 hex digits of that hash also appear in the PNG provenance band, so a paired PNG and JSON can be confirmed to come from the same export click.